DISCRETE MATHEMATICS

CO-WISE QUESTION BANK

B.Tech Computer Engineering | Semester III / IV

Compiled from the Discrete Mathematics question-paper images shared in this chat

Organized by CO1 to CO6 • Repeated questions consolidated • Questions only (no solutions)

Note: Questions are transcribed from the available images. Some questions that depend on a diagram are described rather than redrawn. Where the original paper's notation or numbers were unclear, the question has been phrased conservatively.

CO1 – Mathematical Logic, Propositions and Proofs

Propositions, equivalence and truth tables

1. Show that $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$.

2. Prove the logical equivalence: $(\neg p \wedge (\neg q \wedge r)) \vee (q \wedge r) \vee (p \wedge r) \equiv r$.

3. Prove $p \leftrightarrow q \equiv (\neg p \vee q) \wedge (\neg q \vee p)$ using a truth table.

4. Check whether $[(p \rightarrow q) \wedge (q \rightarrow r) \wedge p] \rightarrow r$ is a tautology, contradiction or contingency.

5. Check whether $[p \rightarrow (q \rightarrow r)] \rightarrow [(p \rightarrow q) \rightarrow (p \rightarrow r)]$ is a tautology.

6. Obtain the disjunctive normal form (DNF) of $(p \wedge \neg(q \wedge r)) \vee (p \rightarrow q)$.

7. Find the conjunctive normal form (CNF) and disjunctive normal form (DNF) of $(\neg p \vee q) \leftrightarrow (p \wedge q)$.

8. Verify the validity of the argument using a truth table: $p \rightarrow q$, $\neg q$, therefore $\neg p$.

9. Test the validity of the argument: If I study, then I will not fail mathematics. If I do not play basketball, then I will study. But I failed mathematics. Therefore, I played basketball.

10. Determine the validity of the argument: If you invest in the stock market, you will get rich. If you get rich, you will be happy. Therefore, if you invest in the stock market, you will be happy.

11. Are these arguments valid? (i) $p \vee q$, $\neg q$, therefore p. (ii) x is positive or x is negative; in either case $x^2 > 0$; therefore $x^2 > 0$.

12. Show that the following argument is valid: If two sides of a triangle are equal, then the opposite angles are equal. Two sides of a triangle are not equal. Therefore, the opposite angles are not equal.

Predicates and quantifiers

1. If $A = \{1,2,3,4\}$, find the truth values of (i) $(\exists x)(x + 5 = 15)$, (ii) $(\forall x)(x + 3 < 10)$, where $x \in A$.

2. If $A = \{1,2,3,\ldots\}$, find the truth values of (i) $(\forall x)(x + 3 > 2)$, (ii) $(\forall x)(x + 2 > 5)$, where $x \in A$.

Mathematical induction

1. Prove by mathematical induction that $1^3 + 2^3 + \ldots + n^3 = [n^2(n + 1)^2]/4$.

2. Prove by mathematical induction that $1^2 + 2^2 + \ldots + n^2 = n(n + 1)(2n + 1)/6$.

3. Prove by mathematical induction that $1 + 3 + 5 + \ldots + (2n - 1) = n^2$.

4. Prove by mathematical induction that 2^(2n) – 1 is divisible by 3 for $n \in N$.

CO2 – Relations, Functions, Posets and Hasse Diagrams

Relations and equivalence / partial orders

1. A relation R on Z is defined by xRy iff $x < y + 1$. Examine whether R is reflexive, symmetric and transitive.

2. Let R be a relation on Z defined by aRb iff $a - b$ is a non-negative even integer. Verify whether R is a partial order.

3. Let R be a relation on integers defined by aRb iff $a - b$ is even. Determine whether R is an equivalence relation.

4. Let R be a relation on integers such that xRy iff $x - y$ is divisible by 7. Show that R is an equivalence relation.

5. Let R be a relation on real numbers such that aRb iff $|a - b| \leq 7$. Is R an equivalence relation? Justify.

6. Determine whether each relation is reflexive, symmetric and/or transitive: (i) {(1,1),(1,2),(2,1),(2,2),(2,3),(3,2),(3,3),(4,4)}; (ii) {(1,1),(1,3),(3,1),(3,3),(4,4)}.

7. Let A be the set of divisors of 30 and R the relation 'is divisible by'. Obtain the relation matrix and draw the Hasse diagram.

Hasse diagrams and lattices

1. Draw the Hasse diagram of A = {2,3,6,12,24,36,72} under divisibility. Is it a lattice?

2. Construct the Hasse diagram for the divisibility relation on S = {1,2,3,5,6,15,24,30}.

3. Draw the Hasse diagram for the poset D■■■ = {1,3,5,15,21,35,70} under divisibility.

4. If A = {1,2,3,4,6,8,24}, show that aRb iff a divides b is a partial order and draw its Hasse diagram.

5. Consider A = {1,2,3,5,30,60}. Let aRb iff a divides b. Prove that R is a partial order and draw the Hasse diagram.

Functions

1. If f,g: $R \to R$ are defined by $f(x) = 2x + 3$ and $g(x) = 3x - 4$, find f■g and g■f.

2. Let $f(x) = (3x + 5)/(2x - 1)$. Show that f is bijective on the stated domain and find $f■^{1}(x)$.

3. Let $f(x) = (2x + 5)/(3x - 7)$, with the excluded domain and range values given in the paper. Prove f is bijective and compute $f■^{1}(x)$.

4. Show that $f(x) = 1/(x - 2)$, $f: R\backslash\{2\} \to R\backslash\{0\}$, is a bijection and find its inverse.

5. Examine whether $f: R\backslash\{5\} \to R\backslash\{0\}$, defined by $f(x) = 1/(x - 5)$, is bijective.

6. Let $f(x) = (2x + 5)/(3x - 7)$. Prove that f is bijective and find its inverse function.

CO3 – Probability and Counting

Probability

1. From a well-shuffled pack of 52 cards, two cards are drawn one after another without replacement. Find the probability that (i) both are kings, (ii) the first is a king and the second is a queen.

2. A bag contains 7 red and 3 black balls; another bag contains 4 red and 5 black balls. One ball is transferred from the first bag to the second, then a ball is drawn from the second bag. Given that the drawn ball is red, find the probability that a black ball was transferred.

3. The chances that A, B and C become Education Ministers are in the ratio 4:1:2. Their probabilities of introducing reservations in professional colleges are 0.3, 0.8 and 0.5 respectively. Find the probability that the reservation bill will be introduced.

4. Students A, B, C and D solve a problem with probabilities 1/3, 1/2, 1/4 and 1/5 respectively. If all try independently, find the probability that the problem is solved.

5. A disease test is positive with probability 95% if the person has the disease and 5% if they do not. If 0.1% of the population has the disease, find the probability that a person with a positive result actually has the disease.

6. A box contains 5 red and 7 blue balls. Two balls are drawn at random. Find the probability that both are red, given that at least one is red.

7. Four roads lead away from a jail. The probabilities of escape on roads A, B, C and D are 1/8, 1/6, 1/4 and 9/10. If a road is selected at random, find the probability of successful escape.

8. The probabilities that three students A, B and C pass an entrance test are 4/9, 2/9 and 1/3. Their probabilities of admission to the same college are 3/10, 1/2 and 4/5. Find the probability they get admission to the same college.

Counting and pigeonhole principle

1. State the pigeonhole principle. How many people must be in a group to guarantee that at least two people have the same birthday?

CO4 – Graph Theory

Graph basics and graph properties

1. Define a planar graph and a Hamiltonian graph, with examples.

2. Define a Hamiltonian graph and give two examples of Hamiltonian graphs with 5 vertices.

3. Define the terms with examples: (i) simple graph, (ii) directed graph, (iii) bipartite graph.

4. Define Eulerian graph and Hamiltonian graph with examples. For the given graph G, determine whether it is Eulerian and give an Euler circuit if one exists; determine whether it is Hamiltonian and give a Hamiltonian cycle if one exists.

5. Check whether the given pairs of graphs are isomorphic. Explain using vertex degrees and adjacency.

6. Draw at least two non-isomorphic graphs of (i) 3 vertices, (ii) 4 vertices.

7. Define a spanning tree and find all non-isomorphic spanning trees for the given graph K■.

8. Determine whether a simple graph can have 15 vertices each of degree 5. Justify your answer using the handshaking lemma.

9. Check which of the given graphs are Eulerian, Hamiltonian, both or neither.

10. Determine whether the given graph is Eulerian. If it is Eulerian, find an Euler path or Euler circuit.

11. Define integral domain with an example. Give an example of an integral domain which is not a field.

Graph algorithms

1. Solve the travelling-salesman problem for the given weighted graph, starting from v■.

2. Draw the Hasse diagram of the given poset and decide whether it is a lattice.

3. Check and explain whether the two given graphs are isomorphic.

CO5 – Algebraic Structures and Groups

Groups and group properties

1. Prove that the set of cube roots of unity forms a group under multiplication of complex numbers.

2. Prove that {0,1,2,3,4,5} is a finite Abelian group of order 6 under addition modulo 6.

3. Show that G = {1, –1, i, –i} is an Abelian group under complex multiplication.

4. Define the Klein four-group and find the order of each element.

5. Show that (G, +■) is a cyclic group where G = {0,1,2,3,4}. Find its generators.

6. Let G = {1,2,4,7,8,11,13,14} under multiplication modulo 15. Show that G is a group, find the order of all elements, and determine whether it is cyclic.

7. Let G be all invertible 2 × 2 matrices over the real numbers and H the set of matrices with determinant 1. Prove that H is a subgroup of G.

8. Let G = {[a b; c d] | a,b,c,d are integers}. Check whether G is a group under matrix multiplication.

9. Prove that the real numbers form a group under the operation a*b = a + b – 2.

10. On Z define a*b = a + b + 1. Show that (Z, *) is an Abelian group.

11. Is usual subtraction on Z associative? Give a reason and an example.

Modular arithmetic

1. Consider A = {1,2,3,4,5,6} under multiplication modulo 7. Construct the multiplication table, find the inverses of 2, 3, 5 and 6, and prove that A is a cyclic group.

2. Show that Z■* = {1,2,3,4} is a cyclic group under multiplication modulo 5. Find all generators.

3. Determine whether the set of non-zero residues modulo 7 forms a group under multiplication modulo 7.

CO6 – Coding Theory, Recurrence Relations and Binary Distance

Encoding functions and error detection

1. For the encoding function e: $B^2 \to B^■$, e(00)=0000, e(10)=0110, e(01)=1011, e(11)=1100, show whether e is a group code.

2. For e: $B^2 \to B^■$ given by e(00)=000000, e(01)=011110, e(10)=101010, e(11)=111000: (i) find the minimum distance, (ii) how many errors can it detect?

3. For e: $B^2 \to B^■$ given by e(00)=00000, e(01)=01110, e(10)=10101, e(11)=11011, show that it is a group code.

4. For e: $B^2 \to B^■$ given by e(00)=00000, e(01)=01110, e(10)=10101, e(11)=11111, find the minimum distance and number of errors detectable.

5. For e: $B^2 \to B^■$ given by e(00)=000000, e(01)=011110, e(10)=101010, e(11)=111000, find the minimum distance of the code and the number of errors it can detect.

6. Find the Hamming weight of the words 11111, 010101 and 0110.

7. Find the Hamming distance between x and y where (i) x=110110, y=000101; (ii) x=001100, y=010110.

8. Determine the (2,5) group code for the given parity-check matrix H = [[1,1,0],[0,1,1],[1,0,0],[0,1,0],[0,0,1]].

Recurrence relations

1. Solve the recurrence relation $a_■ = a_{■■■} + 2$, for $n \geq 2$, with $a_■ = 3$.

2. Solve the recurrence relation $a_■ - 7a_{■■■} + 10a_{■■■} = 0$ with initial conditions $a_■ = 1$ and $a_■ = 6$.